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| **Fiscal Year:** | FY 2020 | **Task Last Updated:** | FY 10/30/2020 |
| **PI Name:** | Weitz, David Ph.D. | | |
| **Project Title:** | Physics of Colloids in Space | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | COMPLEX FLUIDS/SOFT MATTER--Complex Fluids | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 02138 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2009-10 Fluid Physics Rapid Turn Around Flight Opportunity |
| **Start Date:** | 11/01/2013 | **End Date:** | 07/31/2020 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Meyer, William V | **Contact Phone:** | 216-433-5011 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: Extended to 7/31/2020 per NSSC information (Ed., 9/25/19)<br>NOTE: New grant number 80NSSC19K0598 awarded for same work; extended to 11/04/2019 (Ed., 8/1/19)<br>NOTE: Extended to 9/30/2019 per NSSC information (Ed., 10/12/18) | | |
| **Key Personnel Changes/Previous PI:** | September 2019 report: No personnel changes. | | |
| **COI Name (Institution):** | Lu, Peter Ph.D. ( Harvard University ) | | |
| **Grant/Contract No.:** | 80NSSC19K0598 ; NNX13AQ48G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | NOTE: Continues "Physics of Colloids in Space," grant #NNX08AE09G.<br>NOTE (Ed., August 2019): New grant number 80NSSC19K0598 awarded for same work so title now "Physics of Colloids in Space--80NSSC19K0598, NNX13AQ48G"<br><br>We will study the general collective behavior of colloidal particles, including the structures they form and the dynamics of the associated processes. As a result of longterm support from NASA, we have developed significant expertise in the formulation and characterization of systems of hard-sphere colloidal particles in a variety of solvents, and a host of techniques in the laboratory to characterize their structure and behavior, including several forms of microscopy, notably laser-scanning confocal microscopy, optical macro-photography, and laser-light scattering. Using chemistry again developed over the course of many years, supported by NASA, and published in several peer-reviewed publications, we can very carefully control the interaction potential between these colloidal particles, in general making the colloidal particles hard-spheres, interacting only insofar as to repel when contacting each other. Through the mechanism of depletion, we can make the particles attractive, and can exert very precise control over the range and strength of this attraction. We have also developed new sample chamber technology, to improve the optics and mechanics for confocal microscopy, based on design ideas of the current sample chambers on-orbit; the techniques to produce these chambers have been published in peer-reviewed publications. All of these ground-based studies have provided core technologies that we expect to apply to microgravity experiments, in particular the Light Microscopy Module/Advanced Colloids Experiment (LMM/ACE) set of experiments that involve confocal microscopy aboard the International Space Station (ISS). The main goal of our work on colloids is to continue to explore fundamental science through the behavior of these well characterized systems on orbit, with the LMM-confocal, which offers an unprecented 3D view of samples which have evolved in microgravity for years, allowing unprecedented access to long-timescale physics that is simply not possible in any research facility except for ISS.<br><br>The work that we will do for the next several years primarily involves operating our experiments onboard the ISS, namely the ACE family of experiments (ACE-M2, ACE-M2R, follow-up to ACE-M2R with additional confocal, etc.), and then developing the software tools, possibly involving Deep Learning and Artificial Intelligence to process the images and reconstruct the data sets we need. We will conduct further experiments on the ground to support the samples on ISS, and develop relevant technologies for future potential launches of samples to ISS to broaden our understanding of the underlying physics. While our investigations are focused on the fundamental physics of phase separation, kinetic arrest, gelation, syneresis, crystallization, and related processes in microgravity, we anticipate that our experiments will yield insight into a number of materials systems which may be important to human exploration, habitation and spaceflight beyond Low-Earth Orbit, as well as a number of commercial applications on Earth related to long-term stability of products with similar materials components. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | ANNUAL REPORTING SEPTEMBER 2020<br>Our preliminary observations could contribute fundamental understanding to the long-term stability of personal care, food, and pharmaceuticals. Because current theories and, to our knowledge, all experiments on these gel structures presume internal disorder, our unconfirmed preliminary observation of crystalline order may require a re-think of some ideas and theories that provide the foundation for the understanding of the structure and dynamics of these attractive colloidal suspensions. Many theories have been used to predict stability, particularly in different regions of the relevant phase diagrams; our preliminary results may indicate that these theories may not include some fundamentally-important behavior, which has been neither observed nor predicted definitively in this context. Our data may thus give guidance on how to add to or improve many of these theories, which may then predict more accurately the long-term stability of personal care, food, and pharmaceutical products that over the course of their shelf-lives may undergo significant structural rearrangement and / or phase separation.<br><br>Improved understanding of these aging processes are important for a number of Earthbound applications, relating to the overall problem of shelf-life and stability through transport and long-term storage. We emphasize that the new preliminary observations we make are not necessarily tailored for specific stability inferences in real-world products on the market, and as such do not represent an innovative technology; instead, our findings suggest that the existing theories do not provide a complete understanding, and miss some qualitatively different behavior that we see. Finally, our results may have impact on long-duration human spaceflight and habitation beyond low Earth orbit (LEO), as the same physics governs the long-term evolution of personal care products, pharmaceuticals, and foods, which will be necessary to ensure crew comfort and safety on missions that travel beyond LEO.<br><br>ANNUAL REPORTING SEPTEMBER 2019<br><br>These experiments are extremely important in determining the longer term shelflife stability of commercial products, and we plan to continue our work with some of our industrial contacts and colleagues, including at Procter and Gamble, to extend the utility of what we learn to real products. Our studies could also contribute significant understanding to the stability of many products that will likely be used for extended stays in space, particularly the long-term human habitation, exploration and spaceflight beyond low-Earth orbit (LEO). |
| | FINAL REPORTING SEPTEMBER 2020 [ed. note--submitted October 2020]<br>In September 2019, we conducted the ACE-M2R experiment using the confocal microscope capability of the the Light Microscopy Module (LMM) inside the Fluids Integration Rack (FIR) aboard the International Space Station (ISS). We collected several hundred thousand images, representing several terabytes of data. Due to some of the fundamental limitations of operating on ISS, primarily loss-of-signal (LOS) and corruption during the initial downloads of a small fraction of these images, the data arrived as a large mix of folders and directory structures; these must be re-organized by hand, and reconciled with the flight operations logs, to ensure we know for sure which images were collected from which parts of the sample. We believe that we have successfully collected the data sets that we need, so there are no fundamental issues of data loss, but the imaging data are not in any form that is remotely close to usable with automated analysis. The sheer number of images, and total size of the data set, require an enormous amount of hand-sorting and checking of images, which is ongoing, well beyond the end of our funding period.<br><br>ANNUAL REPORTING SEPTEMBER 2019 |

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| **Task Progress:** | We have made significant progress with the on-orbit operations and data collection of our ACE-M2 samples with the recently-launched confocal microscope attached to the Light-Microscopy Module (LMM) in the Fluids Integration Rack (FIR), on-board ISS.<br><br>We launched a number of colloid samples, including colloid-polymer mixtures that undergo phase separation and possibly gelation, as part of the ACE-M2 experiment, more than half a decade ago. The stable environment of the International Space Station has allowed us to let these samples evolve in microgravity, and the return to image the resulting structures formed, after months or years of evolution. We originally viewed the short-term (days to weeks) evolution of our ACE-M2 samples with the LMM using its widefield fluorescence capability, which gives a general overview, but (due to fundamental optical limitations) essentially combines the structure from all layers in the sample into one projected image. The confocal microscope, however, is able to slice through optically the entire sample, and give a 3D picture of the structure.<br><br>In 2018, we were able to image our macroscopic well of a colloid-polymer mixture which appears to have formed a kinetically-arrested gel phase. We collected around 200 full-3D image volumes, mapping out our entire mm-scale sample well with micron-scale resolution. In 2019, we are scheduled to run the next iteration of the experiment, with 2-micron particles that should be able to be imaged individually, in 3D, with the existing confocal microscope system.<br><br>We have received the data from our 2018 experiments, and are processing it now, first organizing the hundreds of thousands of images we collected. |
| **Bibliography Type:** | Description: (Last Updated: 06/13/2025) |
| **Articles in Peer-reviewed Journals** | Zheng W, Zhao S, Yin Y, Zhang H, Needham DM, Evans ED, Dai CL, Lu PJ, Alm EJ, Weitz DA. "High-throughput, single-microbe genomics with strain resolution, applied to a human gut microbiome." Science. 2022 Jun 3;376(6597):eabm1483. http://dx.doi.org/10.1126/science.abm1483 ; PMID: 35653470 , Jun-2022 |